1.Problem statement:

The objective of this project is to design and simulate an automated pneumatic sorting system using MATLAB and Simulink. The system must detect incoming objects, classify them into different categories, and direct them to appropriate lanes using pneumatic actuators. The model should accurately represent real-world industrial automation by incorporating sensing, decision-making, timing control, and physical actuation dynamics.

2.Introduction:

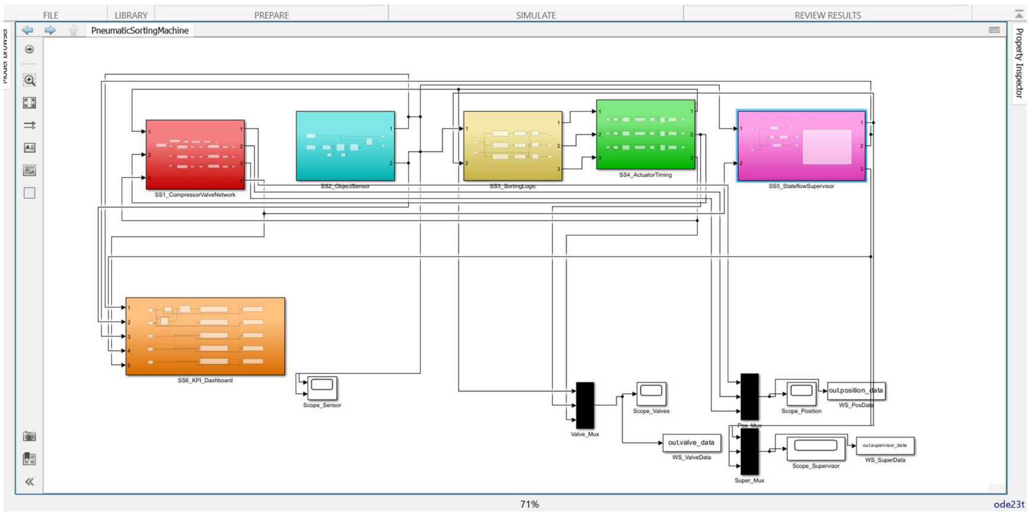
Automation plays a vital role in modern manufacturing industries by improving productivity, accuracy, and safety. One of the key applications of automation is **sorting systems**, which are widely used in packaging, food processing, logistics, and material handling industries. Pneumatic systems, which use compressed air for actuation, are preferred in such applications due to their simplicity, reliability, and fast response.

This project presents the design and simulation of a **Pneumatic Sorting Machine using MATLAB and Simulink**. The system is capable of detecting incoming objects, classifying them into different categories, and directing them to the appropriate lanes using pneumatic actuators. The model integrates multiple subsystems including sensor modeling, decision logic, actuator timing, and pneumatic dynamics to accurately represent real-world industrial operations.

In addition to the basic sorting mechanism, the system incorporates a **Stateflow-based supervisory controller** for managing operational states such as IDLE, RUNNING, FAULT, ESTOP, and RESET. This enhances system safety by detecting faults such as pressure drops and enabling automatic recovery. A **KPI (Key Performance Indicator) Dashboard** is also implemented to monitor important parameters like cycle count, active lane, fault status, system mode, and pressure condition in real time.

Overall, the project demonstrates a **cyber-physical system** that combines control logic, mechanical modeling, and supervisory intelligence, providing a comprehensive simulation of an industrial sorting machine.

3.Model Architecture Diagram:



4.Subsystem / Block Descriptions:

1. Object Sensor Model:

This subsystem simulates object detection using a pulse generator. A random number generator assigns a class (1, 2, or 3) to each detected object. The output consists of a sensor pulse signal and an object classification signal.

2. Sorting Decision Logic:

This subsystem compares the object class with predefined lane values using comparator blocks. Based on the comparison, it generates lane-specific control signals. Boolean outputs are converted to double precision to ensure compatibility with downstream blocks.

3. Actuator Timing Control:

This subsystem ensures that actuator signals remain active for a fixed duration (dwell time). An integrator-based timing mechanism holds the signal high for sufficient time, ensuring proper actuation.

4. Compressor & Valve Network:

This subsystem models the pneumatic system. It includes:

- Valve dynamics (first-order system)
- Pressure scaling (6 bar)

- Force calculation (F = P × A)
- Cylinder dynamics (second-order system)
- Stroke limitation using saturation

The output represents the physical displacement of the actuator.

5. Solver Selection Justification:

The solver used is ode23t (Trapezoidal Rule).

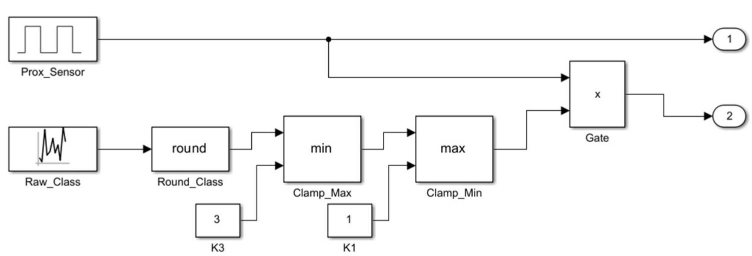
Justification:

- Suitable for stiff systems with moderate nonlinearity
- Efficient for continuous-time models
- Provides stable results for mechanical and pneumatic systems
- Handles transfer functions and integrators effectively

A variable-step solver is used to improve accuracy and reduce computation time.

6.Screenshots of Subsystem Logic:

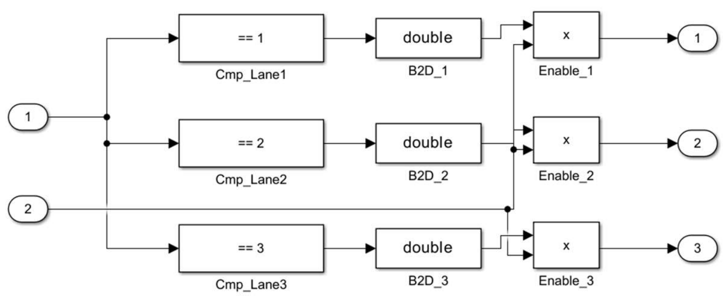
ObjectSensor:



- Simulates detection of objects using a proximity sensor.
- Generates a pulse signal whenever an object is present.
- Uses a random number generator to assign object classes (1, 2, or 3).
- Applies rounding and clamping to ensure valid class values.

- Multiplies sensor pulse with class value to produce valid classification only when object is detected.
- Outputs:
  - Sensor signal (0/1)
  - Object class (1, 2, or 3)

SortingLogic:



- Receives the object class as input.
- Uses comparison blocks to check which lane the object belongs to.
- Implements logic:

  Class = 1 → Lane 1

  Class = 2 → Lane 2

  Class = 3 → Lane 3

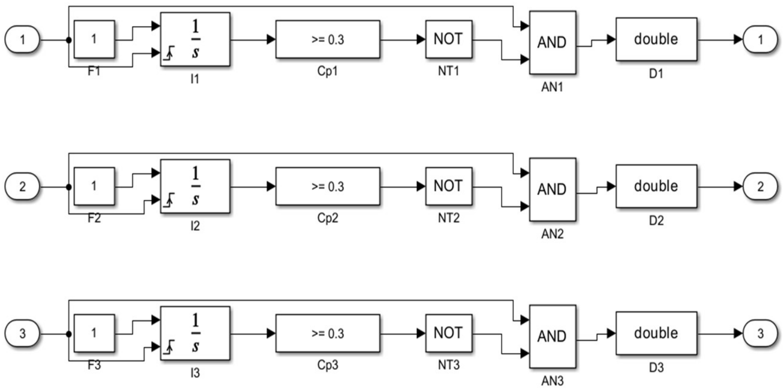
- Converts logical outputs (true/false) into numeric signals (0/1).
- Generates separate control signals for each lane.
- Outputs:

  Lane 1 command

  Lane 2 command

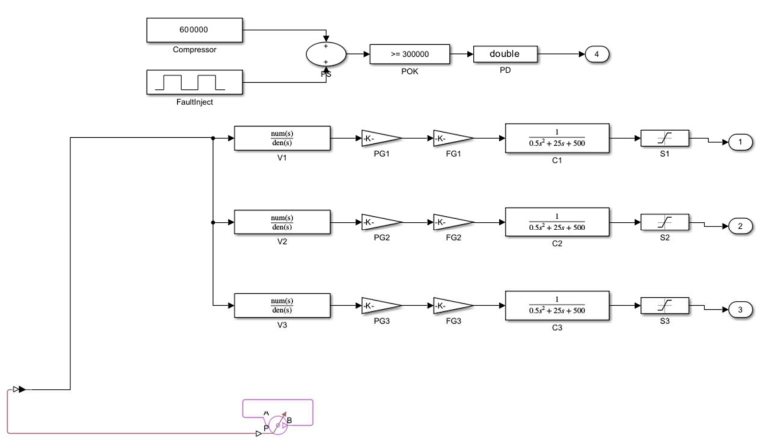
  Lane 3 command

Actuator Timing:



- Ensures that each actuator is activated for a fixed duration (dwell time).
- Uses an integrator as a timer to measure how long the signal is active.
- Compares elapsed time with predefined dwell time (e.g., 0.3 sec).
- Uses logic gates (NOT, AND) to control signal duration.
- Prevents premature deactivation of actuators.
- Converts logical signals to double type for compatibility.
- Outputs:
  - Timed valve signals for each lane

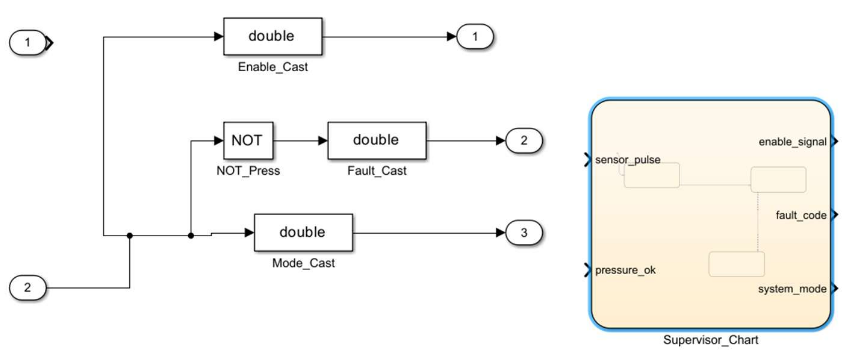
CompressorValveNetwork:

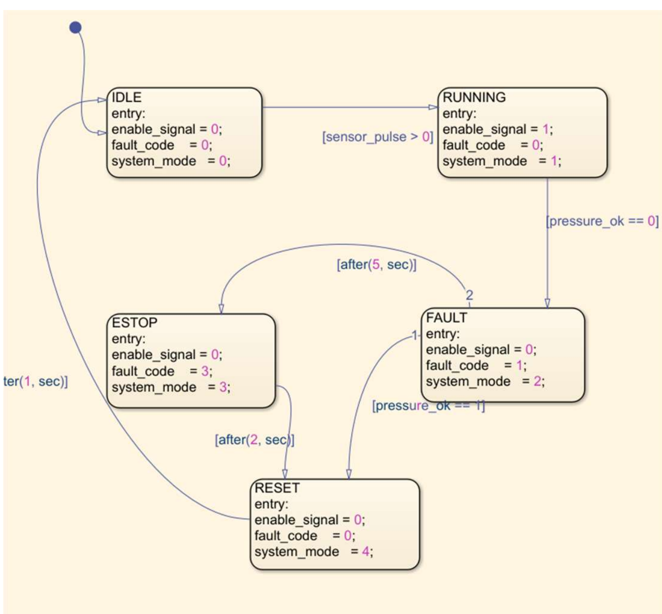


- Compressor provides 6 bar air pressure.

- Simscape block for the valve is added to give pressure to valve blocks
- Valve transfer function models valve opening delay.
- Pressure gain converts signal into air pressure.
- Force gain converts pressure into piston force.
- Cylinder transfer function models piston movement.
- Saturation block limits maximum stroke to 0.15 m.
- Fault injection block simulates pressure failure.

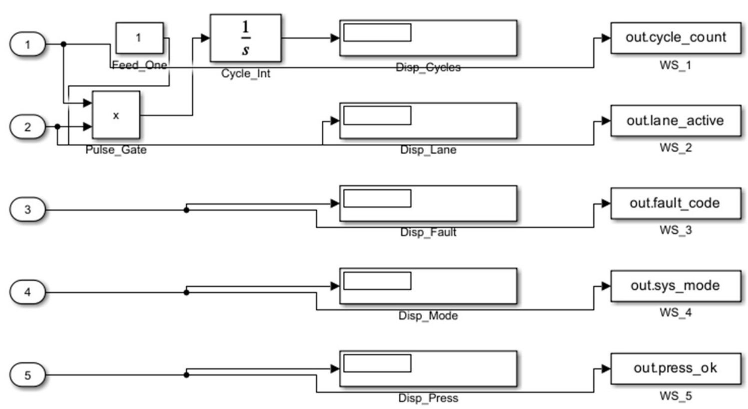
StateflowSupervisor:





- IDLE state waits for object detection.
- RUNNING state enables sorting operation.
- FAULT state activates when pressure drops.
- ESTOP state performs emergency stop.
- RESET state prepares system restart.
- Transitions occur based on sensor and pressure conditions.
- Stateflow improves safety and fault handling.

KPI Dashboard:

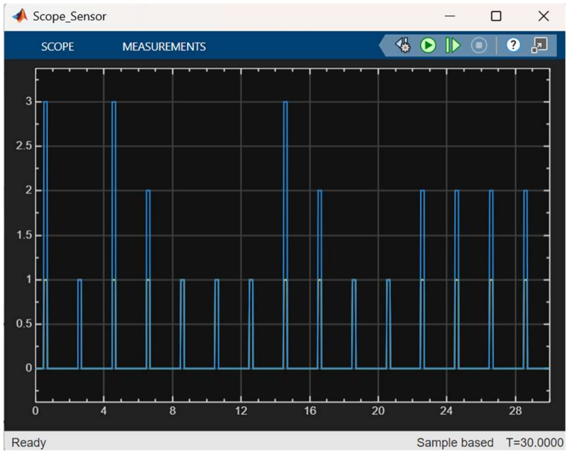


- Enables **real-time monitoring** of system performance
- Helps in **fault diagnosis and debugging**
- Provides **quantitative data for analysis**
- Useful for **performance evaluation and optimization**
- Mimics real industrial **Human-Machine Interface (HMI)**

5. Simulation Results & Plots:

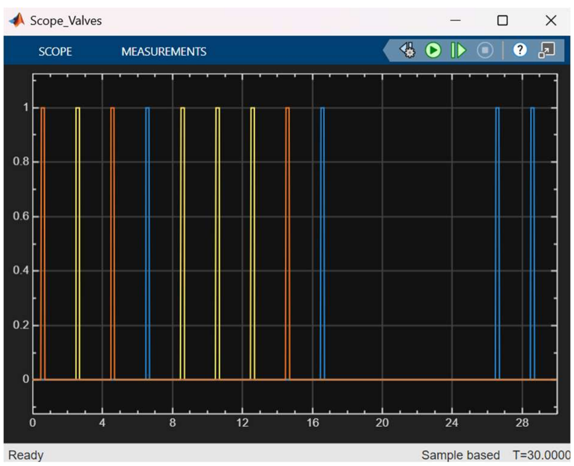
The simulation produces three main outputs:

1. Sensor Output



- Displays object detection pulses
- Shows object classification values (1, 2, 3)

2. Valve Signals



Yellow pulse

Blue pulse

Orange pulse

These represent:

Lane 1 valve

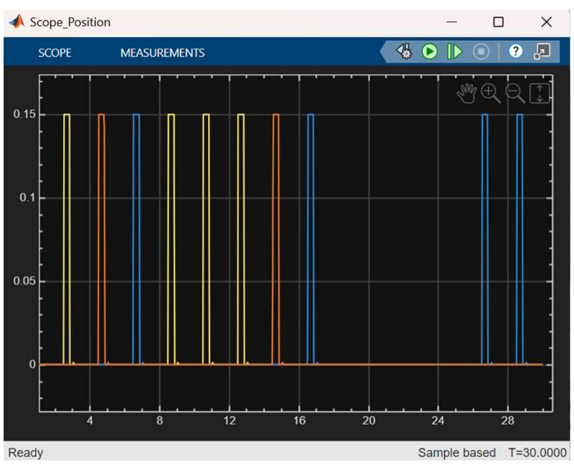
Lane 2 valve

Lane 3 valve

- Indicates which lane is activated
- Only one lane is active at a time
- Signals are extended due to dwell time control

Time:	State:	Valve Output:
0–18 s	RUNNING	Pulses present
18–23 s	FAULT	No pulses
23–25 s	ESTOP	No pulses
25–26 s	RESET	No pulses
27 s onward	RUNNING	Pulses resume

3. Cylinder Position



- Shows actuator displacement
- Maximum stroke reaches 0.15 m
- Saturation ensures physical limits are not exceeded

6. OBSERVATIONS , VALIDATION , DISCUSSIONS:

Observations

- The sensor pulse appears periodically, indicating the arrival of objects at regular intervals.

- The object class signal varies between 1, 2, and 3, representing different categories of objects.
- The sorting logic correctly activates only one lane corresponding to the object class at any given time.
- The actuator timing control maintains the valve signal for a fixed duration (dwell time), even after the input pulse disappears.
- The valve output signals are smooth and show a slight delay due to the transfer function dynamics.
- The cylinder position gradually increases when activated and returns to zero after the signal ends.
- The stroke saturation limits ensure that the cylinder does not exceed the maximum displacement.
- No overlap is observed between lane signals, ensuring proper sorting without conflict.

Validation

The system behaviour matches the expected logical condition:

- Class 1 → Lane 1 activated
- Class 2 → Lane 2 activated
- Class 3 → Lane 3 activated

The timing subsystem is validated by observing that:

- Output remains active for the specified dwell time (~0.3 sec).
- The signal is independent of the input pulse width after triggering.

The pneumatic model is validated by:

- Smooth rise and fall of cylinder position (realistic physical response).
- No instantaneous jumps, confirming dynamic system behavior.

The saturation block validation:

- Cylinder displacement remains within the defined limit (0 to stroke length).

The overall system validation:

- Input (sensor + class) correctly produces expected output (sorted lane + actuator movement).
- No unexpected or undefined outputs are observed during simulation.

Discussion

The pneumatic sorting system successfully integrates sensing, decision logic, timing control, and actuation to achieve automated sorting. The sorting logic accurately directs objects to the correct lane based on their classification without signal overlap.

The actuator timing ensures reliable operation by maintaining valve activation for a fixed duration. The pneumatic model provides realistic behaviour with smooth cylinder motion and controlled displacement within limits.

7. STUDENT REFLECTIONS:

- The project demonstrates effective integration of sensor, logic, timing, and pneumatic actuation in a single system.
- The modular design using subsystems improves clarity, scalability, and ease of debugging.
- The use of transfer functions provides a realistic representation of physical behavior instead of ideal responses.
- Timing control plays a crucial role in ensuring reliable actuator performance.
- Minor limitations include dependence on assumed parameters and lack of real hardware validation.

8. CONCLUSION

The pneumatic sorting machine model successfully simulates an automated sorting process using Simulink. The system accurately detects objects, classifies them, and directs them to appropriate lanes using pneumatic actuators. The integration of decision logic and timing control ensures smooth and reliable operation. The simulation results validate the effectiveness of the model in representing real industrial sorting systems. Overall, the project provides a strong foundation for understanding automation and pneumatic control systems.

9. REFERENCES:

- MATLAB & Simulink Documentation
- Pneumatic Systems Fundamentals
- Control Systems Engineering
- Industrial Automation Basics
- Simulink Modeling Tutorials (Online)

10. APPENDICES:

- **Appendix A — Simulink Onramp Certificate**

- **Appendix B — Stateflow Onramp Certificate**

- **Appendix C — Simscape Onramp Certificate**

MathWorks | Training Services

Course Completion Certificate

Jahnavi Krishna

has successfully completed **100%** of the self-paced training course

Simscape Onramp

DIRECTOR, TRAINING SERVICES

21 February 2026

- **Appendix D---Matlab Onramp Certificate**

MathWorks | Training Services

Course Completion Certificate

Jahnavi Krishna

has successfully completed **100%** of the self-paced training course

MATLAB Onramp

DIRECTOR, TRAINING SERVICES

22 February 2026

- **Appendix E— Model Advisory report**

Model Advisor Report – PneumaticSortingMachine.slx

Simulink version: 25.2 **Model version:** 1.7

System: PneumaticSortingMachine **Current run:** 03-May-2026 19:39:12

Treat as Referenced Model: off

Configuration File:

Justifications File:

Run Summary

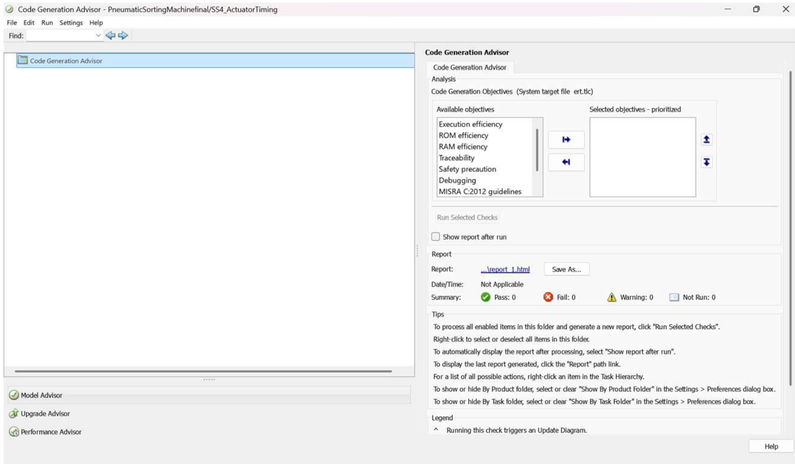
Incomplete	0
Failed	31
Warning	288
Justified	0
Information	0
Passed	625
Not Run	1345
Total	2289

Model Advisor

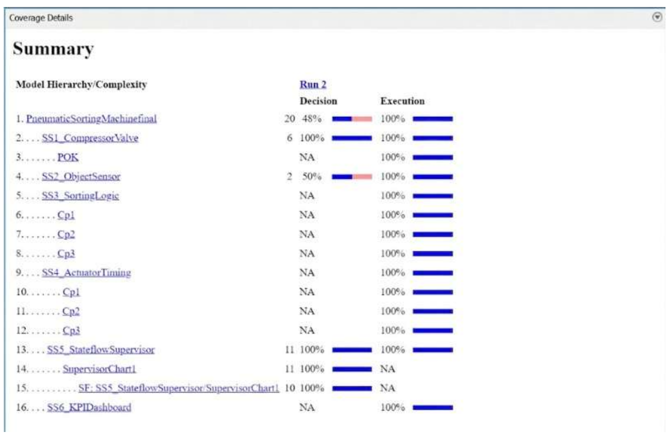
By Product 0 0 0 0 0 0 893

Simulink 0 0 0 0 0 0 44

- **Appendix F – code generation report**



- **Appendix G – coverage report**


Coverage Details

Summary

Model Hierarchy/Complexity		Run 2 Decision	Execution
1. PneumaticSortingMachinefinal	20	48%	100%
2. . . . SS1_CompressorValve	6	100%	100%
3. . . . . . . POK		NA	100%
4. . . . SS2_ObjectSensor	2	50%	100%
5. . . . SS3_SortingLogic		NA	100%
6. . . . . . . Cp1		NA	100%
7. . . . . . . Cp2		NA	100%
8. . . . . . . Cp3		NA	100%
9. . . . SS4_ActuatorTiming		NA	100%
10. . . . . . . Cp1		NA	100%
11. . . . . . . Cp2		NA	100%
12. . . . . . . Cp3		NA	100%
13. . . . SS5_StateflowSupervisor	11	100%	100%
14. . . . . . . SupervisorChart1	11	100%	NA
15. . . . . SF: SS5_StateflowSupervisor/SupervisorChart1	10	100%	NA
16. . . . SS6_KPIDashboard		NA	100%